



Lenovo NVIDIA GB300 NVL72 System Configuration Guide



Machine Type: 7DJV

Note

Before using this information and the product it supports, be sure to read and understand the safety information and the safety instructions, which are available at:

https://pubs.lenovo.com/safety_documentation/

In addition, be sure that you are familiar with the terms and conditions of the Lenovo warranty for your server, which can be found at:

<http://datacentersupport.lenovo.com/warrantylookup>

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Safety

Before installing this product, read the Safety Information.

قبل تركيب هذا المنتج، يجب قراءة الملاحظات الأمنية

Antes de instalar este produto, leia as Informações de Segurança.

在安裝本产品之前，请仔细阅读 Safety Information（安全信息）。

安裝本產品之前，請先閱讀「安全資訊」。

Prije instalacije ovog produkta obavezno pročitajte Sigurnosne Upute.

Před instalací tohoto produktu si přečtěte příručku bezpečnostních instrukcí.

Læs sikkerhedsforskrifterne, før du installerer dette produkt.

Lees voordat u dit product installeert eerst de veiligheidsvoorschriften.

Ennen kuin asennat tämän tuotteen, lue turvaohjeet kohdasta Safety Information.

Avant d'installer ce produit, lisez les consignes de sécurité.

Vor der Installation dieses Produkts die Sicherheitshinweise lesen.

Πριν εγκαταστήσετε το προϊόν αυτό, διαβάστε τις πληροφορίες ασφάλειας (safety information).

לפני שתתקינו מוצר זה, קראו את הוראות הבטיחות.

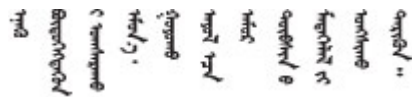
A termék telepítése előtt olvassa el a Biztonsági előírásokat!

Prima di installare questo prodotto, leggere le Informazioni sulla Sicurezza.

製品の設置の前に、安全情報をお読みください。

본 제품을 설치하기 전에 안전 정보를 읽으십시오.

Пред да се инсталира овој продукт, прочитајте информацијата за безбедност.



Les sikkerhetsinformasjonen (Safety Information) før du installerer dette produktet.

Przed zainstalowaniem tego produktu, należy zapoznać się z książką "Informacje dotyczące bezpieczeństwa" (Safety Information).

Antes de instalar este produto, leia as Informações sobre Segurança.

Перед установкой продукта прочтите инструкции по технике безопасности.

Pred inštaláciou tohto zariadenia si pečítajte Bezpečnostné predpisy.

Pred namestitvijo tega proizvoda preberite Varnostne informacije.

Antes de instalar este producto, lea la información de seguridad.

Läs säkerhetsinformationen innan du installerar den här produkten.

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བྱ་འདྲ་མིན་ཡིད་པའི་འོད་ཟེར་བལྟ་དགོས།

Bu ürünü kurmadan önce güvenlik bilgilerini okuyun.

مەزكۇر مەھسۇلاتنى ئورنىتىشتىن بۇرۇن بىخەتەرلىك ئۇچۇرلىرىنى ئوقۇپ چىقىڭ.

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canjbinj soengq cungj vahgangj ancien siusik.

Safety inspection checklist

Use the information in this section to identify potentially unsafe conditions with your rack solution. As each machine was designed and built, required safety items were installed to protect users and service technicians from injury.

Note: The product is not suitable for use at visual display workplaces according to §2 of the Workplace Regulations.

Note: The set-up of the rack solution is made in the server room only.

CAUTION:

This equipment must be serviced by trained personnel, as defined by the IEC 62368-1, the standard for Safety of Electronic Equipment within the Field of Audio/Video, Information Technology and Communication Technology. Lenovo assumes you are qualified in the servicing of equipment and trained in recognizing hazards energy levels in products. Equipment must be installed in a restricted access location and access to the equipment is controlled by the authority responsible for the location.

Important: Electrical grounding of the rack solution is required for operator safety and correct system function. Proper grounding of the electrical outlet can be verified by a certified electrician.

Use the following checklist to verify that there are no potentially unsafe conditions:

1. Make sure that the power is off and the power cord is disconnected.
2. Check the power cord.
 - Make sure that the third-wire ground connector is in good condition. Use a meter to measure third-wire ground continuity for 0.1 ohm or less between the external ground pin and the frame ground.
 - Make sure that the power cord is the correct type.

To view the power cords that are available for the rack solution:

- a. Go to:

<http://dcsc.lenovo.com/#/>

- b. Click **Preconfigured Model** or **Configure to order**.
- c. Enter the machine type and model for your rack solution to display the configurator page.
- d. Click **Power → Power Cables** to see all line cords.
 - Make sure that the insulation is not frayed or worn.
3. Check for any obvious non-Lenovo alterations. Use good judgment as to the safety of any non-Lenovo alterations.
4. Check inside the rack solution for any obvious unsafe conditions, such as metal filings, contamination, water or other liquid, or signs of fire or smoke damage.
5. Check for worn, frayed, or pinched cables.
6. Make sure that the power-supply cover fasteners (screws or rivets) have not been removed or tampered with.

Chapter 1. Introduction

The Lenovo NVIDIA GB300 NVL72 (Type 7DJV) is a revolutionary, liquid-cooled rack-scale solution that integrates seventy-two high-performance Blackwell Ultra GPUs and thirty-six Grace processors into a single, unified system. This liquid-cooled powerhouse represents the pinnacle of Generative AI infrastructure, featuring the NVLink Switch System which enables all seventy-two GPUs to communicate as a single, massive autonomous engine. With high-speed PCIe 6.0 connectivity and fifth-generation NVLink technology, the solution provides unprecedented bandwidth and ultra-low latency data transfer across the entire fabric. Designed to solve the world's most complex challenges, it is engineered to accelerate trillion-parameter AI model training, real-time inference, and massive-scale scientific simulations.

Figure 1. Lenovo NVIDIA GB300 NVL72



Features

Performance, operational efficiency, and fabric reliability were the primary considerations in the design of the rack solution.

The rack solution implements the following features and technologies:

- **High HBM3e bandwidth**

The rack solution utilizes HBM3e technology to deliver a total of 20.7 TB of HBM3e across the 72-GPU NVLink domain. This allows the system to process large AI models instantly and handle massive scientific data.

- **High network bandwidth**

The rack solution features high-speed networking via NVIDIA BlueField-3 DPU adapters or ConnectX-8 SuperNICs. This integrated architecture provides massive aggregate bandwidth across the entire rack, ensuring seamless data flow for large-scale AI workloads.

- **Large system-memory capacity**

The system utilizes a unified memory architecture featuring LPDDR5X memory with error correcting code (ECC) integrated directly onto the NVIDIA Grace processor. This provides up to 17.3 TB of system memory per rack.

- **Large data-storage capacity and hot-swap capability**

Each compute tray supports high-speed E1.S drives, providing localized data staging. With the hot-swap feature, you can add, remove, or replace solid-state drives without turning off the compute tray.

- **Lightpath Diagnostics**

Lightpath Diagnostics provides LEDs to help you diagnose problems. For more information about the Lightpath Diagnostics, see [“Troubleshooting by system LEDs and diagnostics display” on page 23](#).

Tech Tips

Lenovo continually updates the support website with the latest tips and techniques that you can use to solve issues that your rack solution might encounter. These Tech Tips (also called retain tips or service bulletins) provide procedures to work around issues or solve problems related to the operation of your rack solution.

To find the Tech Tips available for your rack solution:

1. Go to <http://datacentersupport.lenovo.com> and navigate to the support page for your rack solution.
2. Click on **How To's** from the navigation pane.
3. Click **Article Type → Solution** from the drop-down menu.

Follow the on-screen instructions to choose the category for the problem that you are having.

Security advisories

Lenovo is committed to developing products and services that adhere to the highest security standards in order to protect our customers and their data. When potential vulnerabilities are reported, it is the responsibility of the Lenovo Product Security Incident Response Team (PSIRT) to investigate and provide information to our customers so they may put mitigation plans in place as we work toward providing solutions.

The list of current advisories is available at the following site:

https://datacentersupport.lenovo.com/product_security/home

Specifications

Summary of the features and specifications of the rack solution. Depending on the model, some features might not be available, or some specifications might not apply.

Refer to the below table for specifications categories and the content of each category.

Specification category	Technical specifications	Mechanical specifications	Environmental specifications
Content	• Compute tray • Management switch • NVLink switch tray • Power shelf	• Dimension • Weight	• Acoustic levels (idle/working mode) • Environmental • Water requirements

Technical specifications

Summary of the technical specifications of rack solution. Depending on the model, some features might not be available, or some specifications might not apply.

Compute tray

- **Form factor:** 1U
 - **Processor:**
 - Two NVIDIA Grace™ CPU Superchip
 - Up to 144 Arm Neoverse V2 cores per tray
 - TDP up to 300W
 - **GPU:** Four NVIDIA Blackwell Ultra B300 Tensor Core GPUs with 1.15 TB HBM3e memory per tray
 - **Memory:** 960 GB on-board LPDDR5x memory with error correcting code (ECC) support
 - **Storage:**
 - External: Up to eight 4 TB PCIe 5.0 x4 E1.S hot-swap drives
 - Internal: One 2 TB PCIe 4.0 M.2 NVMe drive
 - **Integrated functions and I/O connectors:**
 - One Micro-USB management port
 - One Mini DisplayPort connector
 - One USB 3.0 connector
 - One RJ-45 BMC management port
 - One RJ-45 OS management port
 - One RJ-45 OOB management port for BlueField-3 DPU adapter (1GbE)
 - Two 400 Gb/s QSFP connectors
 - Four 800 Gb/s OSFP connectors
 - Integrated diagnostics panel
 - NVLink status LED
 - Power button/LED
 - Reset button
 - System ID button/LED
 - System error LED
- Note:** The maximum video resolution is 1920 x 1200 at 60 Hz.
- **Network:**
 - Four 800 Gb/s OSFP connectors with integrated ConnectX-8 SuperNICs
 - Two 400 Gb/s QSFP connectors via BlueField-3 DPU adapter
 - **System cooling:**
 - Two liquid cooling loops
 - Eight 40 mm x 56 mm dual-rotor fans
 - **Supported and certified operating system:** Ubuntu 24.04 with NVIDIA optimized HWE kernel (kopt)
- References:
- Complete list of available operating systems: <https://lenovopress.lenovo.com/osig>.
 - OS deployment instructions, see “Deploy the operating system” on page 44.

Management switch

- **Form factor:** 1U
- **Processor:** Intel x86 Dual-core
- **Memory:** 8 GB DDR4 SO-DIMM with error correcting code (ECC) support
- **Storage:** 20 GB SSD M.2 PCIe Gen 3
- **Switching capacity:** 448 Gb/s
- **Integrated functions and I/O connectors:**
 - One RJ-45 Ethernet management port
 - One RJ-45 console port
 - One USB 2.0 Type-A connector
 - Forty-eight RJ-45 1G Base-T management ports
 - Four 100 GbE QSFP28 ports
 - Integrated diagnostics panel
 - Power supply LED
 - Reset button
 - System ID button/LED
 - System fan status LED
 - System status LED
- **System fan:** Four hot-swap fan modules with N+1 redundancy
- **Supported and certified operating system:** NVIDIA Cumulus Linux

NVLink switch tray

- **Form factor:** 1U
- **Switching capacity:** 14.4 TB/s via fifth-generation NVLink
- **Integrated functions and I/O connectors:**
 - One USB 2.0 Type-A connector
 - One RJ-45 BMC management port
 - One RJ-45 console port
 - One RJ-45 leak detection interface
 - Two RJ-45 Ethernet management ports
 - Integrated diagnostics panel
 - Power button/LED
 - NVLink status LEDs
 - Reset button
 - System ID button/LED
 - System status LED

Power shelf
• Form factor: 1U • Input voltage: – 3-phase Delta (4-wire): 180 – 305 VAC – 3-phase Wye (5-wire): 311 – 528 VAC • Input current: 60 A • Output power: Six 5.5kW power supply units can provide up to 33kW of power • Output voltage: 50 V • Output current: Up to 660 A

Mechanical specifications

Summary of the mechanical specifications of rack solution. Depending on the model, some features might not be available, or some specifications might not apply.

Dimension
Rack solution: • Width: 600 mm • Depth: 1,068 mm (Expands to 1,200 mm with extension frame) • Height: 2,236 mm

Weight
Rack solution: Approximately 1,580 kg (3,483.3 lbs), depending on the configuration

Environmental specifications

Summary of the environmental specifications of rack solution. Depending on the model, some features might not be available, or some specifications might not apply.

Acoustic levels (idle/working mode)
• Fan control from BMC with multiple temperature input • Sound pressure level (LpAm): up to 82.9 dB • Estimated that the rack will operate around 92dBA from rack rear, typical ambient and rack exercised

Environment

- Air temperature:
 - Operating: ASHRAE Class A2 Compliance
 - Temperature range: 10°C to 35°C (50°F to 95°F)
 - Maximum rate of change: ≤ 20°C per hour
 - Altitude de-rating: The maximum allowable operating temperature is de-rated by 1°C for every 300 m (1,000 ft) up to an altitude of 1,371 m (4,500 ft) above sea level.
 - Non-Operating:
 - Standard storage: -40°C to 70°C (Rack is shipped with pressurized nitrogen)
 - With Di-H20: 5°C to 70°C (41°F to 158°F)
 - With PG25: 10°C to 70°C (14°F to 158°F)
- Maximum altitude: 1,524 m (5,000 ft)
- Relative Humidity (non-condensing):
 - Operating: 20% - 80%
 - Humidity transition rate: < 10% per hour
- Particulate contamination

Attention: Airborne particulates and reactive gases acting alone or in combination with other environmental factors such as humidity or temperature might pose a risk to the rack solution. For information about the limits for particulates and gases, see [“Particulate contamination” on page 7](#).

Note: The rack solution is designed for standard data center environment and recommended to be placed in industrial data center.

Water requirements

Refer to the table below for the direct water-cooling standards:

Fluid supply temperature	Required rack flow	Rack pressure drop
25°C	59 LPM	2.3 psi
30°C	71 LPM	3.2 psi
35°C	89 LPM	4.9 psi
40°C	119 LPM	8.5 psi
45°C	177 LPM	18.4 psi

Standard specifications of the NVLink switch tray.

- Operating water temperature: 2°C to 50°C (35.6°F to 122°F) (ASHRAE W45 compliant)

Particulate contamination

Attention: Airborne particulates (including metal flakes or particles) and reactive gases acting alone or in combination with other environmental factors such as humidity or temperature might pose a risk to the device that is described in this document.

Risks that are posed by the presence of excessive particulate levels or concentrations of harmful gases include damage that might cause the device to malfunction or cease functioning altogether. This specification sets forth limits for particulates and gases that are intended to avoid such damage. The limits must not be viewed or used as definitive limits, because numerous other factors, such as temperature or moisture content of the air, can influence the impact of particulates or environmental corrosives and gaseous contaminant transfer. In the absence of specific limits that are set forth in this document, you must implement practices that maintain particulate and gas levels that are consistent with the protection of human health and safety. If Lenovo determines that the levels of particulates or gases in your environment have caused damage to the device, Lenovo may condition provision of repair or replacement of devices or parts

on implementation of appropriate remedial measures to mitigate such environmental contamination. Implementation of such remedial measures is a customer responsibility.

Table 1. Limits for particulates and gases

Contaminant	Limits
Reactive gases	Severity level G1 as per ANSI/ISA 71.04-1985¹: - The copper reactivity level shall be less than 200 Angstroms per month ($\text{\AA}/\text{month} \approx 0.0035 \mu\text{g}/\text{cm}^2\text{-hour weight gain}$).² - The silver reactivity level shall be less than 200 Angstroms per month ($\text{\AA}/\text{month} \approx 0.0035 \mu\text{g}/\text{cm}^2\text{-hour weight gain}$).³ - The reactive monitoring of gaseous corrosivity must be conducted approximately 5 cm (2 in.) in front of the rack on the air inlet side at one-quarter and three-quarter frame height off the floor or where the air velocity is much higher.
Airborne particulates	Data centers must meet the cleanliness level of ISO 14644-1 class 8. For data centers without airside economizer, the ISO 14644-1 class 8 cleanliness might be met by choosing one of the following filtration methods: - The room air might be continuously filtered with MERV 8 filters. - Air entering a data center might be filtered with MERV 11 or preferably MERV 13 filters. For data centers with airside economizers, the choice of filters to achieve ISO class 8 cleanliness depends on the specific conditions present at that data center. - The deliquescent relative humidity of the particulate contamination should be more than 60% RH.⁴ - Data centers must be free of zinc whiskers.⁵
¹ ANSI/ISA-71.04-1985. <i>Environmental conditions for process measurement and control systems: Airborne contaminants</i>. Instrument Society of America, Research Triangle Park, North Carolina, U.S.A. ² The derivation of the equivalence between the rate of copper corrosion growth in the thickness of the corrosion product in $\text{\AA}/\text{month}$ and the rate of weight gain assumes that Cu_2S and Cu_2O grow in equal proportions. ³ The derivation of the equivalence between the rate of silver corrosion growth in the thickness of the corrosion product in $\text{\AA}/\text{month}$ and the rate of weight gain assumes that Ag_2S is the only corrosion product. ⁴ The deliquescent relative humidity of particulate contamination is the relative humidity at which the dust absorbs enough water to become wet and promote ionic conduction. ⁵ Surface debris is randomly collected from 10 areas of the data center on a 1.5 cm diameter disk of sticky electrically conductive tape on a metal stub. If examination of the sticky tape in a scanning electron microscope reveals no zinc whiskers, the data center is considered free of zinc whiskers.	

Management options

Every device is equipped with a dedicated management interface.

Component	Management interface	Access method	Function
Compute tray	Host BMC	- Web GUI - Redfish API	- Monitoring of the system inventory - System configuration change (e.g., BMC network setting) - System network setting - Firmware updates - Service data collection

Component	Management interface	Access method	Function
	GPU management	NVIDIA System Management Interface (nvidia-smi)	In-band monitoring and management of the GPUs
Management switch	NVIDIA Cumulus Linux	• NVUE CLI • Redfish API	Network configuration
NVLink switch tray	• NVIDIA Cumulus Linux • Switch BMC	• NVUE CLI • Redfish API • Redfish CLI	• Monitoring of the system inventory and firmware version • Firmware updates • System log collection
Power shelf	Power Management Controller	• Web GUI • Redfish CLI	• Firmware updates • Monitoring of the system inventory and health status • Power management • System log collection

Chapter 2. Rack solution components

This section contains information about each of the components associated with the rack solution.

- “Compute tray components” on page 11
- “Management switch components” on page 14
- “NVLink switch tray components” on page 16
- “Power shelf components” on page 18
- “Rack components” on page 20

Compute tray components

This section contains information on the compute tray.

Compute tray front view

This section contains information on the front view of the compute tray.

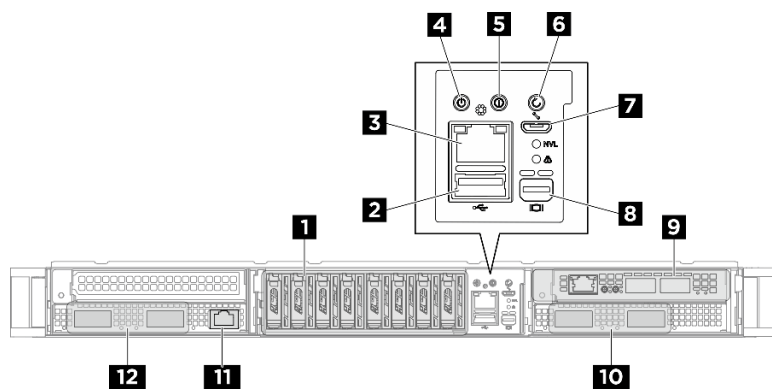


Figure 2. Compute tray front view

Table 2. Components on the compute tray front view

1 E1.S hot-swap drive bays (0 to 7)	2 USB 3.0 connector
3 RJ-45 BMC management port	4 Power button/LED
5 System ID button/LED	6 Reset Button
7 Micro-USB management port	8 Mini DisplayPort connector
9 DPU adapter (BlueField-3 B3240)	10 OSFP card
11 RJ-45 OS management port	12 OSFP card

1 E1.S hot-swap drive bays (0 to 7)

Install E1.S drives to these bays. See “Install an E1.S hot-swap drive to a compute tray” in *User Guide* or *Hardware Maintenance Guide*.

2 USB 3.0 connector

Connect a USB device to this connector.

3 RJ-45 BMC management port

Provides dedicated access to the Baseboard Management Controller (BMC) for remote monitoring, power control, and firmware management.

4 Power button/LED

Press this button to turn the system on and off manually. The states of the LED are as follows:

Status	Color	Description
Off	None	The system is off.
Blinking (1 HZ)	Green	The system is in power-off state.
Blinking (4 HZ)	Green	The system is running POST.
Solid on	Green	The system is on and running.

5 System ID button/LED

Use this system ID button and the system ID LED to visually locate the system. The states of the LED are as follows:

Status	Color	Description
Off	None	The system is in a normal operating state.
Blinking (1 HZ)	Blue	The location identification feature is activated.

6 Reset button

Press and hold the reset button with a flat object to reset the system.

7 Micro-USB management port

Provides a CPU UART interface for debugging and diagnostics.

8 Mini DisplayPort connector

Connect a monitor to this connector.

Note: The maximum video resolution is 1920 x 1200 at 60 Hz.

9 DPU adapter (BlueField-3 B3240)

An NVIDIA BlueField-3 B3240 DPU adapter provides high-performance network acceleration and offload capabilities. It is equipped with one 1GbE RJ-45 OOB management port and two QSFP ports supporting up to 400 Gb/s Ethernet or NDR InfiniBand connectivity.

10 OSFP card

Each card is equipped with two 800 Gb/s OSFP connectors, supporting direct GPU connections for high-speed data transfers.

11 RJ-45 OS management port

Provides a dedicated 1GbE interface that serves for the Host OS network connection.

12 OSFP card

Each card is equipped with two 800 Gb/s OSFP connectors, supporting direct GPU connections for high-speed data transfers.

Compute tray rear view

This section contains information on the rear view of the compute tray.

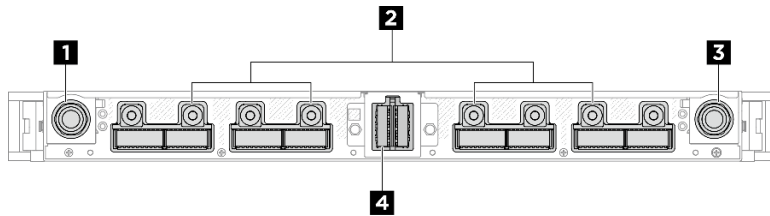


Figure 3. Compute tray rear view

Table 3. Components on the compute tray rear view

1 Coolant return	2 Cable cartridge connectors
3 Coolant supply	4 Busbar connector

1 Coolant return

The exit point that discharges heated coolant out of the system.

2 Cable cartridge connectors

Connect the system to the cable cartridges.

3 Coolant supply

The entry point that receives chilled coolant into the system.

4 Busbar connector

Connects the system to the busbar.

Compute tray top view

This section contains information on the top view of the compute tray.

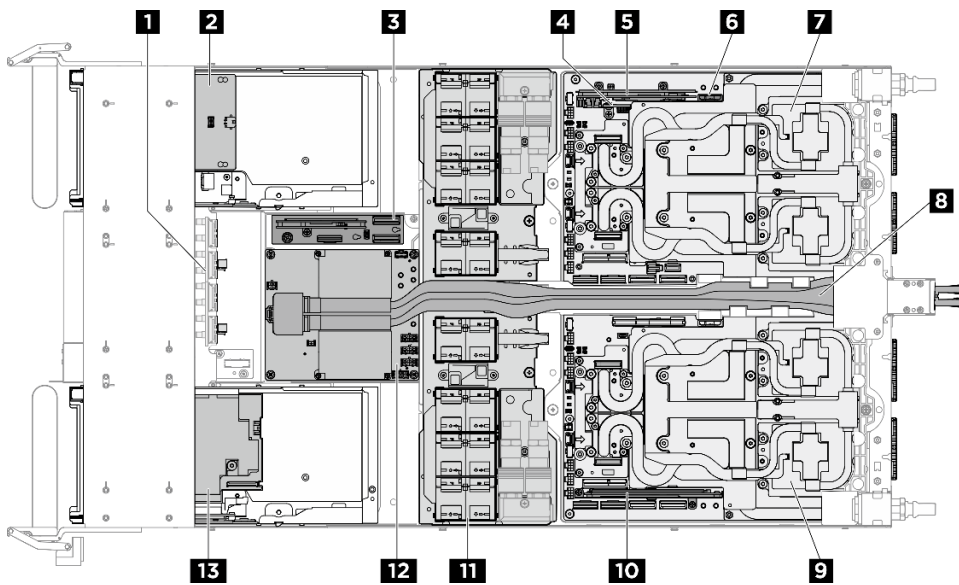


Figure 4. Compute tray top view

Table 4. Components on the compute tray top view

1 E1.S drive backplane	2 OSFP card
3 BMC card	4 TPM card
5 HMC card	6 CMOS battery (CR2032)
7 Compute board with cold plate (primary)	8 Busbar cable
9 Compute board with cold plate (secondary)	10 M.2 drive assembly
11 Fans	12 Power distribution board
13 DPU adapter (BlueField-3 B3240)	

Management switch components

This section contains information on the management switch.

Management switch front view

This section contains information on the front view of the management switch.

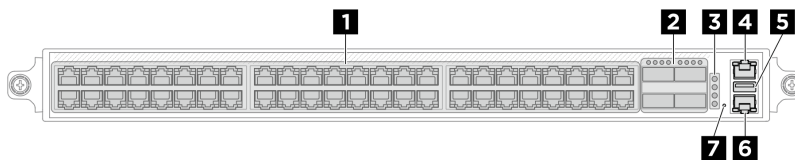


Figure 5. Management switch front view

Table 5. Components on the management switch front view

1 RJ-45 management ports	2 QSFP28 ports
3 System status LEDs	4 RJ-45 Ethernet management port

Table 5. Components on the management switch front view (continued)

5 USB 2.0 Type-A connector	6 RJ-45 console port
7 Reset button	

1 RJ-45 management ports

The management switch includes forty-eight dedicated 1G Base-T RJ-45 ports for Out-of-Band (OOB) management.

2 QSFP28 ports

The management switch includes four 100GbE QSFP28 ports that function as high-speed uplinks. Each port can be split into two 50GbE ports or up to four 25GbE ports using the appropriate breakout cables.

3 System status LEDs

For more information about system status LEDs, see [“Management switch LEDs” on page 25](#).

4 RJ-45 Ethernet management port

The RJ-45 Ethernet management port provides dedicated access for remote management. By default, the port is configured for auto-negotiation, supporting speeds from 100MbE to 1GbE.

5 USB 2.0 Type-A connector

Connect a USB 2.0 device to this connector.

6 RJ-45 console port

This port is intended for initial system configuration and debugging.

7 Reset button

Press and hold the reset button with a flat object to reset the system.

Management switch rear view

This section contains information on the rear view of the management switch.

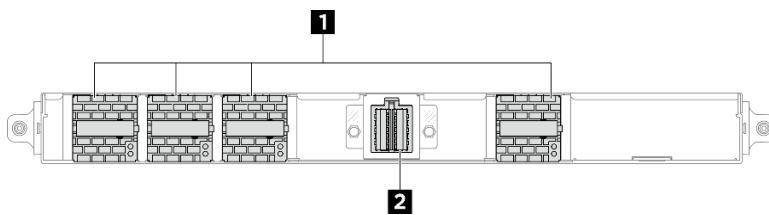


Figure 6. Management switch rear view

Table 6. Components on the management switch rear view

1 Fans	2 Busbar connector
---------------	---------------------------

1 Fans

Four hot-swap fans configured with N+1 redundancy.

2 Busbar connector

Connects the system to the busbar.

NVLink switch tray components

This section contains information on the NVLink switch tray.

NVLink switch tray front view

This section contains information on the front view of the NVLink switch tray.

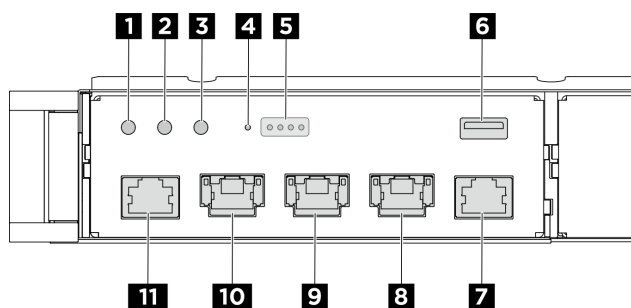


Figure 7. NVLink switch tray front view

Table 7. Components on the NVLink switch tray front view

1 Power button/LED	2 Reset button
3 System ID button/LED	4 System status LED
5 NVLink status LEDs	6 USB 2.0 Type-A connector
7 RJ-45 leak detection interface	8 RJ-45 Ethernet management port 0
9 RJ-45 Ethernet management port 1	10 RJ-45 BMC management port
11 RJ-45 console port	

1 Power button/LED

Press this button to turn the system on and off manually. The states of the LED are as follows:

Status	Color	Description
Off	None	The System is off.
Solid on	Green	The System is on and running.

2 Reset button

Press and hold the reset button with a flat object to reset the system.

3 System ID button/LED

Use this system ID button and the system ID LED to visually locate the system. The states of the LED are as follows:

Status	Color	Description
Off	None	The system is in a normal operating state.
Solid on	Blue	The location identification feature is activated.

4 System status LED

The system status LED helps you to determine if there are any system errors. The states of the LED are as follows:

Status	Color	Description
Blinking	Green	The system is booting or resetting.
Solid on	Green	The system is working correctly.
Solid on	Amber	An error has been detected.

5 NVLink status LEDs

Four LEDs (1, 2, 3, 4) indicating the NVLink switch tray connection status for cartridges 1 through 4. The states of the LED are as follows:

Status	Color	Description
Off	None	The link is down.
Blinking (1 HZ)	Amber	Beaconing (locating) is active on any one of the NVLink ports.
Blinking (2 HZ)	Amber	A link error has been detected.
Solid on	Amber	The physical link is established.
Blinking (1 HZ)	Green	The logical link is established and active.
Solid on	Green	The logical link is established with no active traffic.

6 USB 2.0 Type-A connector

Connect a USB 2.0 device to this connector.

7 RJ-45 leak detection interface

Connect a leak detection system to this interface to monitor for liquid exposure.

8 / 9 RJ-45 Ethernet management port 0 / RJ-45 Ethernet management port 1

These two 1GbE ports are dedicated to the out-of-band (OOB) management of the NVLink switch tray.

10 RJ-45 BMC management port

This 1GbE port is intended for BMC management.

11 RJ-45 console port

This port is intended for initial system configuration and debugging.

NVLink switch tray rear view

This section contains information on the rear view of the NVLink switch tray.

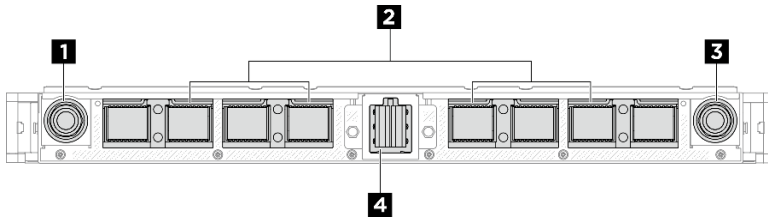


Figure 8. NVLink switch tray rear view

Table 8. Components on the NVLink switch tray rear view

1 Coolant return	2 Cable cartridge connectors
3 Coolant supply	4 Busbar connector

1 Coolant return

The exit point that discharges heated coolant out of the system.

2 Cable cartridge connectors

Connect the system to the cable cartridges.

3 Coolant supply

The entry point that receives chilled coolant into the system.

4 Busbar connector

Connects the system to the busbar.

Power shelf components

This section contains information on the power shelf.

Power shelf front view

This section contains information on the front view of the power shelf.

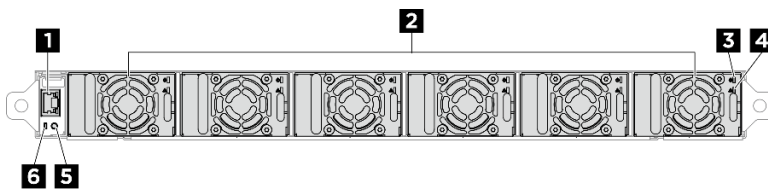


Figure 9. Power shelf front view

Table 9. Components on the power shelf front view

1 PMC module	2 Power supply units
3 PSU power status LED	4 PSU health status LED
5 PMC module reset button	6 PMC module status LED

1 PMC module

The PMC (Power Management Controller) module features a front-facing RJ-45 port that supports PoE (Power over Ethernet) functionality.

2 Power supply units

Six 5.5kW power supply units can provide up to 33 kW of power.

3 PSU power status LED

This LED indicates the power status of each power supply unit. The states of the LED are as follows:

Status	Color	Description
Off	None	The output voltage is off.
Blinking	Green	AC input is normal, but the voltage output is off.
Solid on	Green	The output voltage is active.

4 PSU health status LED

This LED indicates the health status of each power supply unit. The states of the LED are as follows:

Status	Color	Description
Off	Green	The output voltage is off.
Blinking (4 HZ)	Green	AC input is normal, but the voltage output is off.
Solid on	Green	The output voltage is active.
Off	Amber	The power supply unit is operating normally without faults.
Blinking (4 HZ)	Amber	The power supply unit is in bootloader mode.
Solid on	Amber	A fault has been detected.

5 PMC module reset button

Press and hold the reset button with a flat object to reset the PMC module.

6 PMC module status LED

Use this LED to monitor the status of the PMC module. The states of the LED are as follows:

Status	Color	Description
Blinking	Green	The PMC module is currently performing a firmware update.
Solid on	Green	The PMC module is operating normally.
Solid on	Amber	Over-temperature warning.

Power shelf rear view

This section contains information on the rear view of the power shelf.

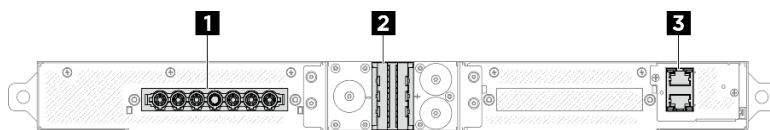


Figure 10. Power shelf rear view

Table 10. Components on the power shelf rear view

1 Power whip connector	2 Busbar connector
3 RJ-45 ports	

1 Power whip connector

Connects the power whip cable to this connector.

2 Busbar connector

Connects the system to the busbar.

3 RJ-45 ports

These two RJ-45 ports support daisy-chaining of multiple power shelves to enable current sharing across the rack.

Rack components

This section contains information on the rack.

Rack front view

This section contains information on the rack front view.

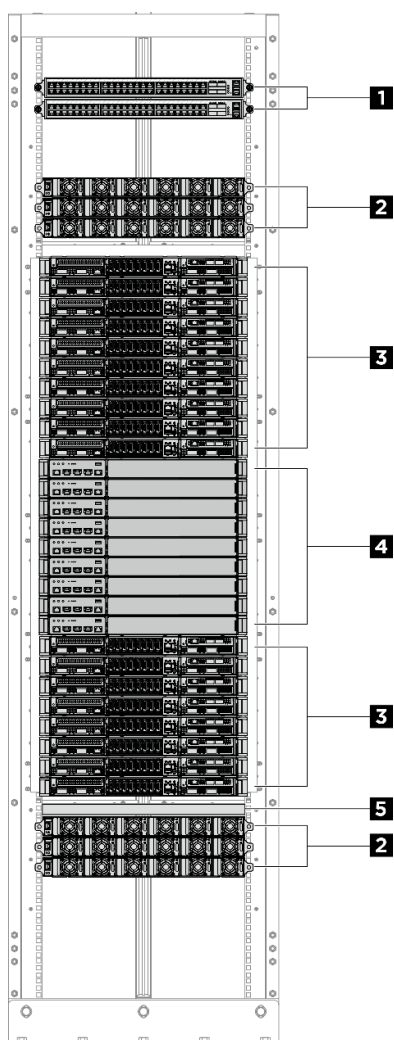


Figure 11. Rack front view

Table 11. Components on the rack front view

1 Management switches	2 Power shelves
3 Compute trays	4 NVLink switch trays
5 Leakage drip pan	

1 Management switches

The rack system includes two management switches, which provide dedicated out-of-band (OOB) connectivity for all compute trays and NVLink switch trays.

2 Power shelves

The rack system is equipped with either six or eight power shelves. Each shelf is capable of delivering up to 33kW of power, providing the high-density energy required to support the full rack configuration.

3 Compute trays

The rack system features eighteen compute trays, providing the foundation for its massive parallel processing capability.

4 NVLink switch trays

The rack system incorporates nine NVLink switch trays to manage high-speed data fabric. All NVLink interconnections are established through rear-side, blind-mate connectors, which interface directly with the cable cartridges in the rack.

5 Leakage drip pan

The rack system includes one leakage drip pan designated to catch drips to prevent damage.

Rack rear view

This section contains information on the rack rear view.

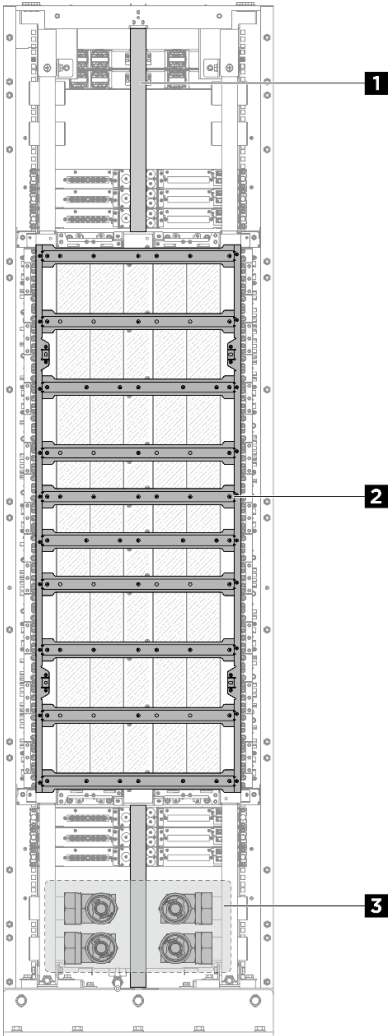


Figure 12. Rack rear view

Table 12. Components on the rack rear view

1 Busbar	2 Cable cartridges
3 Manifold connectors	

1 Busbar

Provides power distribution and connectivity throughout the rack.

2 Cable cartridges

Provide the high-speed physical interface required to unify compute trays and NVLink switch trays.

3 Manifold connectors

Provide the interface for coolant distribution throughout the rack to ensure efficient heat dissipation.

System LEDs and diagnostics display

See the following section for information on available system LEDs and diagnostics display.

For more information, refer to [“Troubleshooting by system LEDs and diagnostics display” on page 23](#).

Troubleshooting by system LEDs and diagnostics display

See the following section for information on available system LEDs and diagnostics display.

Compute tray LEDs

This topic provides information on LEDs of compute tray.

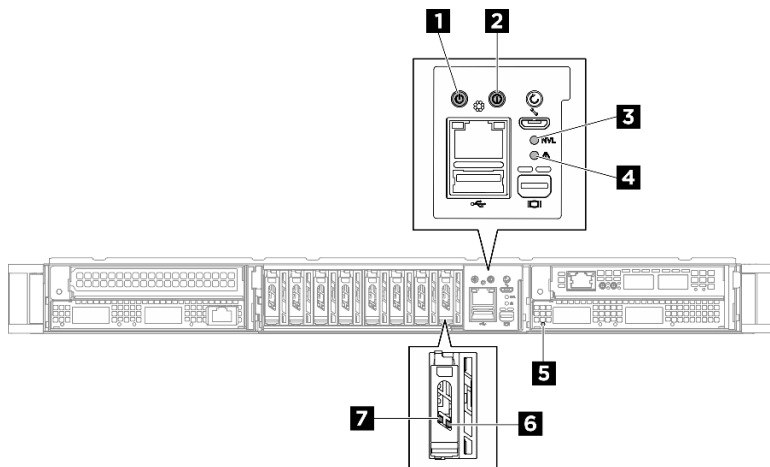


Figure 13. Compute tray LEDs

Table 13. Compute tray LEDs

1 Power button/LED	2 System ID button/LED
3 NVLink status LED	4 System fault LED
5 OSFP port LED	6 E1.S drive status LED
7 E1.S drive activity LED	

1 Power button/LED

Press this button to turn the system on and off manually. The states of the LED are as follows:

Status	Color	Description
Off	None	The system is off.
Blinking (1 HZ)	Green	The system is in power-off state.
Blinking (4 HZ)	Green	The system is running POST.
Solid on	Green	The system is on and running.

2 System ID button/LED

Use this system ID button and the system ID LED to visually locate the system. The states of the LED are as follows:

Status	Color	Description
Off	None	The system is in a normal operating state.
Blinking (1 HZ)	Blue	The location identification feature is activated.

3 NVLink status LED

Use this NVLink status LED to monitor the NVLink connection status. The states of the LED are as follows:

Status	Color	Description
Off	None	The link is down.
Blinking (1 HZ)	Amber	Beaconing (locating) is active on any one of the NVLink ports.
Solid on	Amber	The physical link is established.
Blinking (1 HZ)	Green	The logical link is established and active.
Solid on	Green	The logical link is established with no active traffic.

4 System fault LED

The system fault LED helps you to determine if there are any system errors. The states of the LED are as follows:

Status	Color	Description
Off	None	The system is working correctly.
Blinking (4 HZ)	Amber	A critical event has been detected.
Solid on	Amber	A non-recoverable event has been detected.

5 OSFP port LED

Use the bi-color LED to monitor the activity of the OSFP card. The states of the LED are as follows:

Status	Color	Description
Blinking (4 HZ)	Green	The link is operating at the maximum supported speed.
Blinking (2 HZ)	Green	The link is operating below the maximum supported speed.
Solid on	Green	The link is active.
Blinking (4 HZ)	Yellow	An error has occurred to the link.

Status	Color	Description
Blinking (1 HZ)	Yellow	The beacon command is used to locate the OSFP card.
Solid on	Yellow	A physical link is established in InfiniBand mode. This status is not applicable in Ethernet mode.

Note: Using command to light up locating LED is available when transceiver or cable is attached to the OSFP port.

6 E1.S drive status LED

Use the bi-color LED to monitor the status of the E1.S drive. The states of the LED are as follows:

Status	Color	Description
Off	None	The drive is not present or power is unavailable.
Solid on	Blue	The drive is operating normally.
Solid on	Amber	A drive failure has been detected.

7 E1.S drive activity LED

Use the bi-color LED to monitor the activity of the E1.S drive. The states of the LED are as follows:

Status	Color	Description
Off	None	The drive is not present or power is disconnected.
Blinking (4 HZ)	Green	Drive activity is in progress; removal is allowed.
Solid on	Green	The drive is powered and active; removal is allowed.
Blinking (4 HZ)	White	Drive activity is in progress; removal is not allowed.
Solid on	White	The drive is powered and active; removal is not allowed.

Management switch LEDs

This topic provides information on LEDs of management switch.

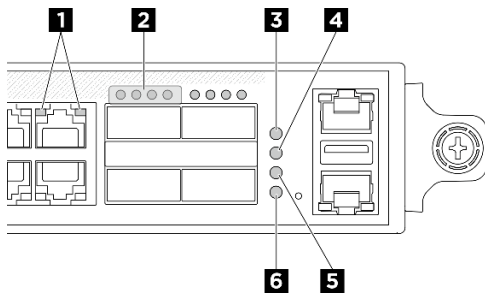


Figure 14. Management switch LEDs

Table 14. Management switch LEDs

1 RJ-45 management port status LEDs	2 QSFP28 port status LEDs
3 System status LED	4 System fan status LED
5 System power status LED	6 System ID button/LED

1 RJ-45 management port status LED

Use these LEDs to monitor the status of the RJ-45 management port. The states of the LEDs are as follows:

Status	Color	Description
Off	None	The Link is down. Ensure that both near-end and far-end connectors are fully seated and verify cable integrity.
Solid on	Yellow	The link is established.
Blinking	Green	The link is connected with active traffic.
Solid on	Green	The link is connected with no active traffic.

2 QSFP28 port status LEDs

Use these LEDs to monitor the status of the QSFP28 port. The states of the LEDs are as follows:

Status	Color	Description
Off	None	The link is down. Check the cable.
Blinking	Yellow	A link error has been detected. Verify cable integrity and replace it if needed.
Blinking	Green	The link is established with active traffic.
Solid on	Green	The link is established with no active traffic.

3 System status LED

Use this LED to monitor the status of the system. The states of the LED are as follows:

Status	Color	Description
Blinking	Green	The System is booting. The boot process may take up to five minutes to complete.
Solid on	Green	The System is powered on and operating normally.
Solid on	Red	A critical error has occurred.

4 System fan status LED

Use this LED to monitor the status of the system fans. The states of the LED are as follows:

Status	Color	Description
Off	None	The System is booting.
Solid on	Green	All fans are operating normally.
Solid on	Red	One or more fans have malfunctioned. Identify and replace the failed fan.

5 System power status LED

Use this LED to monitor the power status of the system. The states of the LED are as follows:

Status	Color	Description
Off	None	AC power loss detected across all power supply units.
Blinking (1 HZ)	Green	AC input is detected, but only the 12VSB is active. The power supply unit is in a Smart-on state.

Status	Color	Description
Solid on	Green	The power supply unit is operating normally.
Blinking	Yellow	The power supply unit continues to operate, but warning events have been detected. These include over-temperature, high power, over-current events, or degraded fan speed.
Solid on	Yellow	- The AC cord is disconnected or AC power has been lost while the second power supply still has AC input power. Ensure the AC cord is securely connected to the inactive unit. - A power supply unit failure has occurred. Verify input voltage.

❶ System ID button/LED

Use this system ID button and the system ID LED to visually locate the system. The states of the LED are as follows:

Status	Color	Description
Off	None	Normal operating state
Solid on	Blue	Location identification feature is activated.

NVLink switch tray LEDs

This topic provides information on LEDs of NVLink switch tray.

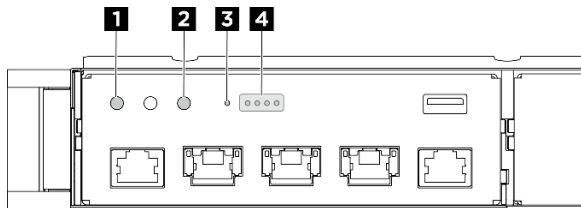


Figure 15. NVLink switch tray LEDs

Table 15. NVLink switch tray LEDs

❶ Power button/LED	❷ System ID button/LED
❸ System status LED	❹ NVLink status LEDs

❶ Power button/LED

Press this button to turn the system on and off manually. The states of the LED are as follows:

Status	Color	Description
Off	None	The System is off.
Solid on	Green	The System is on and running.

❷ System ID button/LED

Use this system ID button and the system ID LED to visually locate the system. The states of the LED are as follows:

Status	Color	Description
Off	None	The system is in a normal operating state.
Solid on	Blue	The location identification feature is activated.

3 System status LED

The system status LED helps you to determine if there are any system errors. The states of the LED are as follows:

Status	Color	Description
Blinking	Green	The system is booting or resetting.
Solid on	Green	The system is working correctly.
Solid on	Amber	An error has been detected.

4 NVLink status LEDs

Four LEDs (1, 2, 3, 4) indicating the NVLink switch tray connection status for cartridges 1 through 4. The states of the LED are as follows:

Status	Color	Description
Off	None	The link is down.
Blinking (1 HZ)	Amber	Beaconing (locating) is active on any one of the NVLink ports.
Blinking (2 HZ)	Amber	A link error has been detected.
Solid on	Amber	The physical link is established.
Blinking (1 HZ)	Green	The logical link is established and active.
Solid on	Green	The logical link is established with no active traffic.

Power shelf LEDs

This topic provides information on LEDs of power shelf.

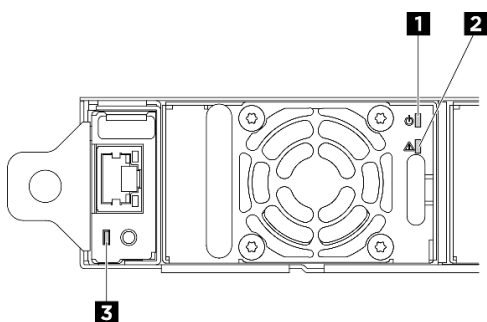


Figure 16. Power shelf LEDs

Table 16. Power shelf LEDs

1 PSU power status LED	2 PSU health status LED
3 PMC module status LED	

1 PSU power status LED

This LED indicates the power status of each power supply unit. The states of the LED are as follows:

Status	Color	Description
Off	None	The output voltage is off.
Blinking	Green	AC input is normal, but the voltage output is off.
Solid on	Green	The output voltage is active.

2 PSU health status LED

This LED indicates the health status of each power supply unit. The states of the LED are as follows:

Status	Color	Description
Off	Green	The output voltage is off.
Blinking (4 HZ)	Green	AC input is normal, but the voltage output is off.
Solid on	Green	The output voltage is active.
Off	Amber	The power supply unit is operating normally without faults.
Blinking (4 HZ)	Amber	The power supply unit is in bootloader mode.
Solid on	Amber	A fault has been detected.

3 PMC module status LED

Use this LED to monitor the status of the PMC module. The states of the LED are as follows:

Status	Color	Description
Blinking	Green	The PMC module is currently performing a firmware update.
Solid on	Green	The PMC module is operating normally.
Solid on	Amber	Over-temperature warning.

Chapter 3. Parts list

Identify each of the components that is available for your rack solution with the parts list.

For more information about ordering parts:

1. Go to <http://datacentersupport.lenovo.com> and navigate to the support page for your rack solution.
2. Click **Parts**.
3. Enter the serial number to view a listing of parts for your rack solution.

It is highly recommended that you check the power summary data for your rack solution using Lenovo Capacity Planner before purchasing any new parts.

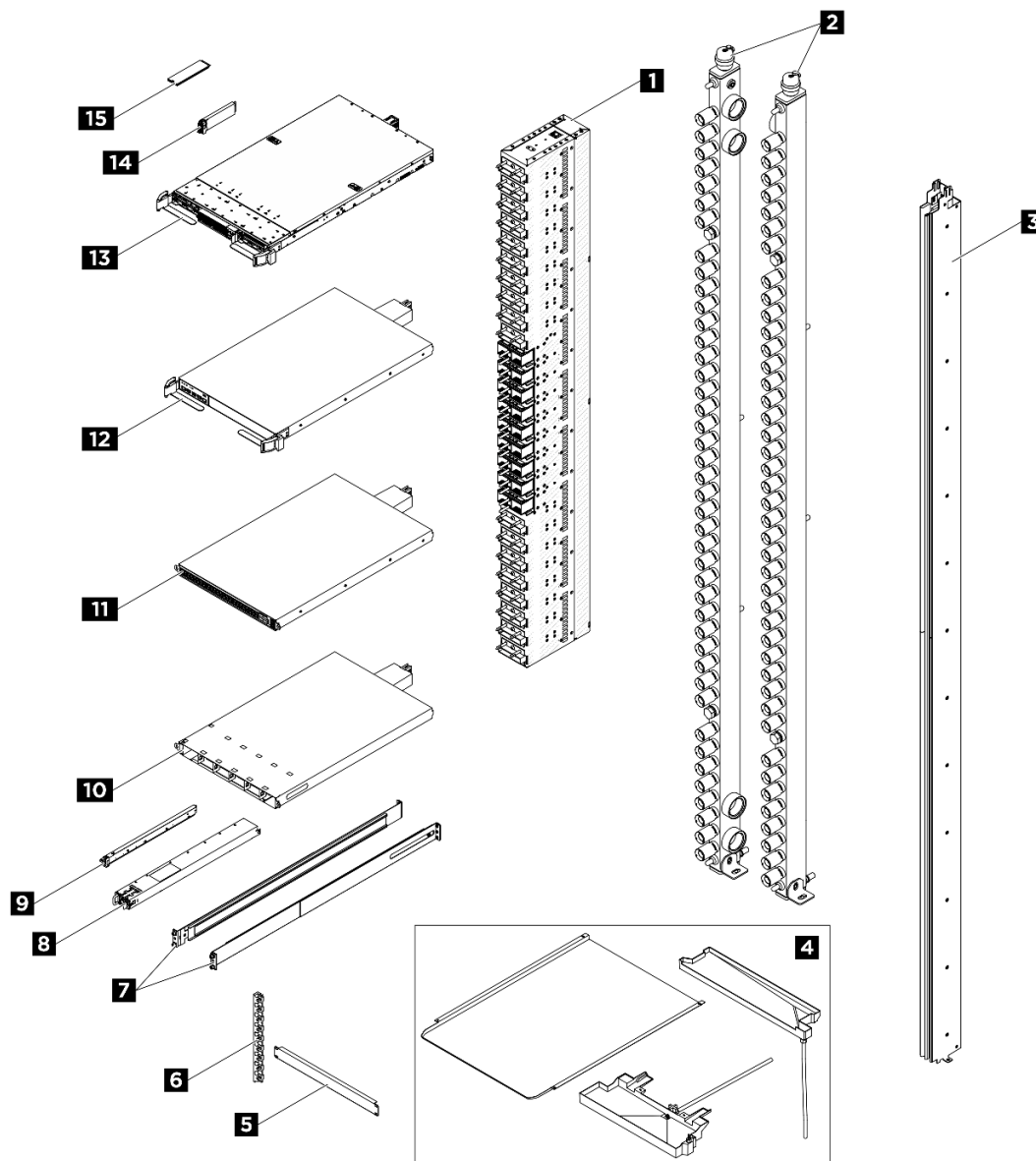


Figure 17. Rack solution components

The parts listed in the following table are identified as one of the following:

- **T1:** Tier 1 customer replaceable unit (CRU). Replacement of Tier 1 CRUs is your responsibility. If Lenovo installs a Tier 1 CRU at your request with no service agreement, you will be charged for the installation.
- **T2:** Tier 2 customer replaceable unit (CRU). You may install a Tier 2 CRU yourself or request Lenovo to install it, at no additional charge, under the type of warranty service that is designated for your rack solution.
- **F:** Field replaceable unit (FRU). FRUs must be installed only by trained service technicians.
- **C:** Consumable and Structural parts. Purchase and replacement of consumable and structural parts (components, such as a filler or bezel) is your responsibility. If Lenovo acquires or installs a structural component at your request, you will be charged for the service.

Table 17. Parts list

Index	Description	Type
For more information about ordering parts: 1. Go to http://datacentersupport.lenovo.com and navigate to the support page for your rack solution. 2. Click Parts. 3. Enter the serial number to view a listing of parts for your rack solution.		
1	Cable cartridge	F
2	Manifold	F
3	Busbar	F
4	Leakage drip pan assembly	F
5	Blank panel	C
6	Ejector bar	C
7	Rail	F
8	Power supply unit	T1
9	PMC module	T1
10	Power shelf	F
11	Management switch	F
12	NVLink switch tray	F
13	Compute tray	F
14	E1.S drive	T1
15	M.2 drive	F

Note: For clarity, specific cable types are not pictured. Refer to the <http://datacentersupport.lenovo.com> for a comprehensive list of replaceable parts and ordering instructions.

Power cords

Several power cords are available, depending on the country and region where the rack solution is installed.

To view the power cords that are available for the rack solution:

1. Go to:

<http://dcsc.lenovo.com/#/>
2. Click **Preconfigured Model** or **Configure to order**.
3. Enter the machine type and model for your rack solution to display the configurator page.
4. Click **Power** → **Power Cables** to see all line cords.

Notes:

- For your safety, a power cord with a grounded attachment plug is provided to use with this product. To avoid electrical shock, always use the power cord and plug with a properly grounded outlet.
- Power cords for this product that are used in the United States and Canada are listed by Underwriter's Laboratories (UL) and certified by the Canadian Standards Association (CSA).

- For units intended to be operated at 115 volts: Use a UL-listed and CSA-certified cord set consisting of a minimum 18 AWG, Type SVT or SJT, three-conductor cord, a maximum of 15 feet in length and a parallel blade, grounding-type attachment plug rated 15 amperes, 125 volts.
- For units intended to be operated at 230 volts (U.S. use): Use a UL-listed and CSA-certified cord set consisting of a minimum 18 AWG, Type SVT or SJT, three-conductor cord, a maximum of 15 feet in length and a tandem blade, grounding-type attachment plug rated 15 amperes, 250 volts.
- For units intended to be operated at 230 volts (outside the U.S.): Use a cord set with a grounding-type attachment plug. The cord set should have the appropriate safety approvals for the country in which the equipment will be installed.
- Power cords for a specific country or region are usually available only in that country or region.

Chapter 4. Unboxing and setup

Information in this section assists you on unboxing and setting up the rack solution. When unboxing the rack solution, check if the items in the package are correct, and learn where to find information of compute tray serial number. Make sure to follow the instructions in [“Rack solution setup checklist” on page 36](#) when setting up the solution.

Rack solution package contents

When you receive your rack solution, verify that the shipment contains everything that you expected to receive.

The rack solution package includes the following items:

- One 48U rack with a busbar, cable cartridges, and manifolds
- Eighteen compute trays
- Two management switches
- Nine NVLink switch trays
- Six or eight power shelves, depending on the configuration
- Material box, including items such as power cords*, accessory kit, and printed documents

Notes:

- Some of the items listed are available on select models only.
- Items marked with asterisk(*) are optional.

If any item is missing or damaged, contact your place of purchase. Ensure that you retain your proof of purchase and packing material. They might be required to receive warranty service.

Identify the compute tray and access the BMC

This section contains instruction on how to identify your compute tray and where to find the BMC access information.

Identifying your compute tray

When you contact Lenovo for help, the machine type and serial number information help support technicians to identify your compute tray and provide faster service.

The illustration below shows the location of the ID label which contains the machine type and serial number of the compute tray.

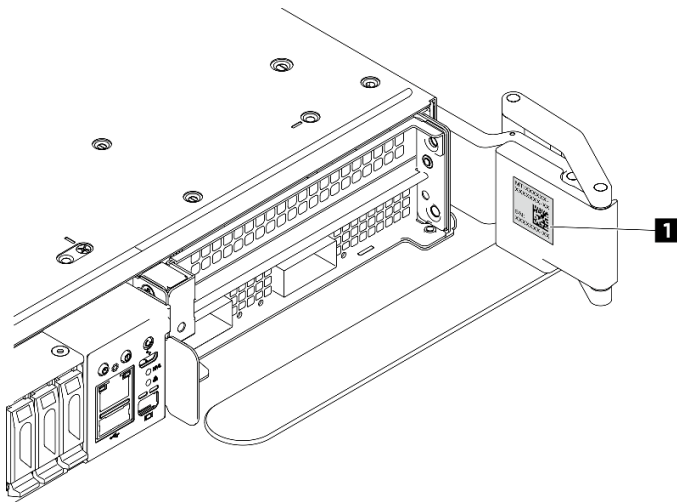


Figure 18. Location of the ID label

MAC address label

The illustration below shows the locations of the MAC address labels for both the system and the BMC.

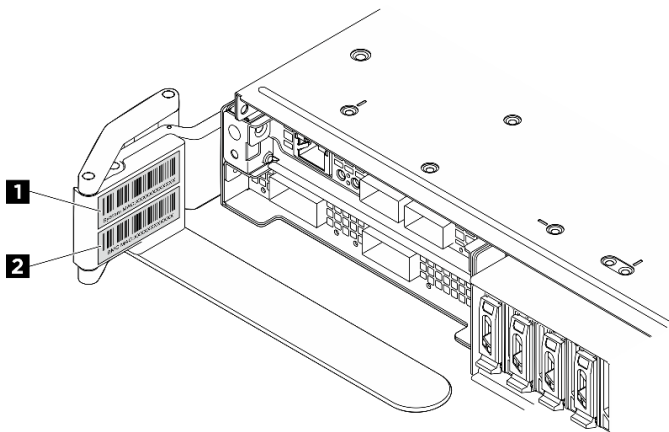


Figure 19. Locations of the MAC address labels

1 System MAC address label	2 BMC MAC address label
-----------------------------------	--------------------------------

Rack solution setup checklist

Use the rack solution setup checklist to ensure that you have performed all tasks that are required to set up your rack solution.

The following steps describe the general procedure for setting up a rack solution.

Setup the rack solution hardware

Complete the following procedures to setup the rack solution hardware.

1. Unpack the rack solution package. See [“Rack solution package contents” on page 35](#).
2. Install any required hardware. See the related topics in “Hardware replacement procedures” in *User Guide* or *Hardware Maintenance Guide*.

3. Purge the air from the rack solution and fill it with coolant.
4. Perform leakage and pressure testing for the rack solution.
5. Connect all external cables to the rack solution. See [Chapter 2 “Rack solution components” on page 11](#) for connectors locations.

Typically, you will need to connect the following cables:

- Connect the rack solution to the power source
- Connect the rack solution to the data network
- Connect the rack solution to the storage device
- Connect the rack solution to the management network

6. Power on the rack solution.

The location of the power button and the status of the power LED for each component are specified in:

- [Chapter 2 “Rack solution components” on page 11](#)
- [“Troubleshooting by system LEDs and diagnostics display” on page 23](#)

Each component in the rack solution can be turned on (power LED on) in any of the following ways:

- You can press the power button on the individual component.
- The rack solution can restart automatically after a power interruption.

7. Validate the rack solution. Make sure that the power LED, Ethernet connector LED, and network LED are lit with green light, which means the rack solution hardware was set up successfully.

See [“Troubleshooting by system LEDs and diagnostics display” on page 23](#) for more information on the LED indications.

Configure the system

Complete the following procedures to configure the system. For detailed instructions, refer to [Chapter 5 “System configuration” on page 39](#).

1. Set the network connection for the BMC to the management network.
2. Update the firmware for the rack solution, if necessary.
3. Configure the firmware for the rack solution.
4. Install the operating system.
5. Back up the rack solution configuration.
6. Install the applications and programs for which the rack solution is intended to be used.

Chapter 5. System configuration

Complete these procedures to configure your system.

Update the firmware

Several options are available to update the firmware for the rack solution.

- The latest firmware can be found at the following site:
 - <https://datacentersupport.lenovo.com/products/solutions-and-software/aiinfrastructure/gb300nvl72/7djb/downloads/driver-list/>
- You can subscribe to product notification to stay up to date on firmware updates:
 - <https://datacentersupport.lenovo.com/solutions/ht509500>

Update Bundles (Service Packs)

Lenovo typically releases firmware in bundles called Update Bundles (Service Packs). To ensure that all of the firmware updates are compatible, you should update all firmware at the same time.

Update method terminology

- **In-band update.** The installation or update is performed using a tool or application within an operating system that is executing on the server's core CPU.
- **Out-of-band update.** The installation or update is performed by the HOST BMC, collecting the update and then directing the update to the target subsystem or device. Out-of-band updates have no dependency on an operating system executing on the core CPU. However, most out-of-band operations do require the server to be in the S0 (Working) power state.
- **On-Target update.** The installation or update is initiated from an installed operating system executing on the target server itself.
- **Update Bundles (Service Packs).** Update Bundles (Service Packs) are bundled updates designed and tested to provide the interdependent level of functionality, performance, and compatibility. Update Bundles (Service Packs) are server machine-type specific and are built (with firmware and device driver updates) to support specific Windows Server, Red Hat Enterprise Linux (RHEL) and SUSE Linux Enterprise Server (SLES) operating system distributions. Machine-type-specific firmware-only Update Bundles (Service Packs) are also available.

Update the compute tray firmware

Use this section to update the firmware for the compute tray and its components.

- [“Update the compute tray firmware” on page 40](#)
- [“Verify the compute tray firmware versions” on page 40](#)
- [“Update the ConnectX-8 I/O board firmware” on page 41](#)
- [“Update the DPU adapter firmware” on page 41](#)
- [“Update the DPU adapter OS and firmware” on page 42](#)
- [“Update the E1.S drive firmware” on page 42](#)
- [“Update the M.2 drive firmware” on page 42](#)

Update the compute tray firmware

- [Update the BMC firmware](#)
- [Update the UEFI firmware](#)
- [Update the HMC firmware](#)

Update the BMC firmware

- Update the compute tray BMC firmware with the following command: `curl --location 'https://<BMC IP address>/redfish/v1/UpdateService/upload' -L -s -k -u '<BMC username>:<BMC password>' --form 'UpdateFile=@<BMC firmware file location and its file name>' --form 'OemParameters="{\"ImageType\": \"BMC\"}";type=application/json' --form 'UpdateParameters="{\"Targets\": [\"/redfish/v1/UpdateService/FirmwareInventory/BMC\"]}";type=application/json' | jq`
- Verify the current BMC firmware version with the following command: `curl -L -s -k -u '<BMC username>:<BMC password>' -X GET "https://<BMC IP address>/redfish/v1/UpdateService/FirmwareInventory/BMCImage1" | jq | grep -A 2 "BMCImage"`

Update the UEFI firmware

- Update the compute tray UEFI firmware with the following command: `curl --location 'https://<BMC IP address>/redfish/v1/UpdateService/upload/' -L -s -k -u '<BMC username>:<BMC password>' --form 'UpdateFile=@<UEFI firmware file location and its file name>' --form 'OemParameters="{\"ImageType\": \"PLDM\", \"Platform\": \"HGX\"}";type=application/json' --form 'UpdateParameters="{\"Targets\": []}";type=application/json' | jq`
- Verify the current UEFI firmware version with the following command: `curl -L -s -k -u '<BMC username>:<BMC password>' -X GET "https://<BMC IP address>/redfish/v1/Systems/System_0" | jq | grep "BiosVersion"`

Update the HMC firmware

Note: Updating the HMC firmware automatically includes updates for the processor and GPU firmware.

- Update the compute tray HMC firmware with the following command: `curl --location 'https://<BMC IP address>/redfish/v1/UpdateService/upload/' -L -s -k -u '<BMC username>:<BMC password>' --form 'UpdateFile=@<HMC firmware file location and its file name>' --form 'OemParameters="{\"ImageType\": \"PLDM\", \"Platform\": \"HGX\"}";type=application/json' --form 'UpdateParameters="{\"Targets\": []}";type=application/json' | jq`
- Verify the current HMC firmware version with the following command: `curl -L -s -k -u '<BMC username>:<BMC password>' -X GET "https://<BMC IP address>/redfish/v1/UpdateService/FirmwareInventory/HGX_FW_BMC_0" | jq | grep "Version"`

Verify the compute tray firmware versions

Use the following commands to verify the firmware versions of the compute tray components: `curl -L -s -k -u '<BMC username>:<BMC password>' -X GET "https://<BMC IP address>/redfish/v1/UpdateService/FirmwareInventory/" | jq`

The output will appear as follows:

Firmware	version
• BMCImage1	0.01.0
• BMCImage2	0.01.0
• E1S_BP_CPLD_0	0b.01.04
• E1S_BP_CPLD_1	0b.01.04
• HGX_FW_BMC_0	GB200Nv1-24.12-9
• HGX_FW_CPLD_0	0.1C
• HGX_FW_CPU_0	00000008
• HGX_FW_CPU_1	00000008
• HGX_FW_ERoT_BMC_0	01.03.0262.0000_n04
• HGX_FW_ERoT_CPU_0	01.03.0262.0000_n04
• HGX_FW_ERoT_CPU_1	01.03.0262.0000_n04
• HGX_FW_ERoT_FPGA_0	01.03.0262.0000_n04
• HGX_FW_ERoT_FPGA_1	01.03.0262.0000_n04
• HGX_FW_FPGA_0	1.1C
• HGX_FW_FPGA_1	1.1C
• HGX_FW_GPU_0	07.00.6C.00.03

Figure 20. Firmware versions of the compute tray components

Update the ConnectX-8 I/O board firmware

Complete the following steps to update the ConnectX-8 I/O board firmware.

1. From the compute tray OS, navigate to the ConnectX-8 folder with the following command: `cd ConnectX-8`
2. Start the NVIDIA MST (Mellanox Software Tools) service with the following command: `sudo mst start`
3. Identify the ConnectX-8 I/O board port names with the following command: `sudo mst status -v`

Note: Ensure all four ports in the compute tray (two per I/O board) are updated.

4. Update the ConnectX-8 I/O board firmware with the following command for each port: `sudo flint -d <ConnectX-8 port name> -i <firmware filename>.bin burn`

Note: Firmware must be updated on each port individually.

5. The system displays the current and new firmware versions followed by a Do you want to continue? (y/n) prompt. Enter `y` to proceed.
6. After the firmware updates are complete for all ports, perform the AC power cycle by running the following command or by using the **Power Control** page in the BMC Web GUI. `sudo ipmitool raw 0x3c 0x01 3 1`
7. After the AC power cycle, use the following command to confirm the updated firmware version within the InfiniBand operational data: `ibstat`

The ConnectX-8 ports are identified in the output as CA (Channel Adapter) type MT4131.

Update the DPU adapter firmware

Complete the following steps to update the firmware for the DPU adapter (BlueField-3 B3240).

1. Navigate to the DPU adapter directory with the following command: `cd BlueField-3`
2. Start the NVIDIA MST (Mellanox Software Tools) service with the following command: `sudo mst start`
3. Identify the DPU adapter port names with the following command: `sudo mst status -v`
4. Update the DPU adapter firmware with the following command: `sudo flint -i <BF-3 DPU port name> -i <firmware filename>.bin burn`

Note: The DPU adapter contains two ports; however, updating only one port applies the firmware update to the entire adapter.

5. The system displays the current and new firmware versions followed by a Do you want to continue? (y/n) prompt. Enter `y` to proceed.

6. After the firmware updates are complete for all ports, perform the AC power cycle by running the following command or by using the **Power Control** page in the BMC Web GUI. `sudo ipmitool raw 0x3c 0x01 3 1`
7. After the AC power cycle, use the following command to confirm the updated firmware version within the InfiniBand operational data: `ibstat`

The DPU adapter ports are identified in the output as CA (Channel Adapter) type MT41692.

Update the DPU adapter OS and firmware

For detailed instructions on updating the OS and firmware for the DPU adapter (BlueField-3 B3240), refer to the official documentation at <https://docs.nvidia.com/networking/index.html>.

Update the E1.S drive firmware

Complete the following steps to update the E1.S drive firmware.

1. Install the NVMe drive management utility with the following command: `sudo apt install nvme-cli`
 2. Identify the current E1.S drive firmware location and version with the following command: `sudo nvme list`
- Note:** E1.S drive firmware names begin with the letter L.
3. Map the new firmware file to the drive with the following command: `sudo nvme fw-download <firmware location> -f <firmware filename>.bin`
 4. Update the E1.S drive firmware with the following command: `sudo nvme fw-commit <firmware location> -s 2 -a 3`
 5. After the firmware updates are complete for all ports, perform the AC power cycle by running the following command or by using the **Power Control** page in the BMC Web GUI. `sudo ipmitool raw 0x3c 0x01 3 1`
 6. After the AC power cycle, confirm the updated firmware version with the following command: `sudo nvme list`

Update the M.2 drive firmware

Complete the following steps to update the M.2 drive firmware.

1. Install the NVMe drive management utility with the following command: `sudo apt install nvme-cli`
 2. Identify the current M.2 drive firmware location and version with the following command: `sudo nvme list`
- Note:** M.2 drive firmware names begin with the letter G.
3. Map the new firmware file to the drive with the following command: `sudo nvme fw-download <firmware location> -f <firmware filename>.bin`
 4. Update the M.2 drive firmware with the following command: `sudo nvme fw-commit <firmware location> -s 2 -a 3`
 5. After the firmware updates are complete for all ports, perform the AC power cycle by running the following command or by using the **Power Control** page in the BMC Web GUI. `sudo ipmitool raw 0x3c 0x01 3 1`
 6. After the AC power cycle, confirm the updated firmware version with the following command: `sudo nvme list`

Update the management switch OS

Use this section to update Cumulus Linux OS for the management switch.

Update Cumulus Linux OS on the management switch using NVUE or Linux commands. For complete instructions and command syntax, refer to <https://docs.nvidia.com/networking-ethernet-software/cumulus-linux/Installation-Management/Upgrading-Cumulus-Linux/>.

Notes:

- Always back up the switch configuration before performing an image upgrade.
- Identify current firmware image levels with the command: `nv show system image`

Update the NVLink switch tray firmware

Use this section to update the firmware for the NVLink switch tray.

Use NVUE commands to update the NVLink switch tray firmware, then reboot the system for the updates to take effect. For complete instructions and command syntax, refer to https://docs.nvidia.com/networking/display/nvidianvosusermanualforNVLinkswitchesv25021884/upgrading+system+firmware#src-3660559829_UpdatingSystemFirmware-UpgradeFirmware.

Note: Identify current firmware image levels with the following command: `nv show platform firmware ASIC`

Update the power shelf firmware

Use this section to update the firmware for the power shelf .

- [“Delta power shelf” on page 43](#)
- [“Lite-On power shelf” on page 43](#)

Delta power shelf

Update the Delta power shelf PMC or PSU firmware using the Linux command `curl` from a Redfish-enabled client system. Both PMC and PSU firmware files are provided in a compressed .tar format.

Note: PSU firmware updates are applied at the shelf level. Initiating a PSU firmware update will apply the update to all six PSUs simultaneously.

Command Syntax: `$ curl -ks -u <PMC username>:<PMC password> -X POST 'https://<PMC IP address>/redfish/v1/UpdateService/update' --data-binary "@<PSU or PMC firmware file path>"`

Lite-On power shelf

- [Update the PMC firmware via Redfish CLI commands](#)
- [Update the PSU firmware via Redfish CLI commands](#)
- [Update the firmware via PMC Web GUI](#)

Update the PMC firmware via Redfish CLI commands

Update the Lite-On power shelf PMC firmware using the Linux command `curl` from a Redfish-enabled client system.

Command Syntax:

```
$ curl -k -u <PMC username>:<PMC password> https://<PMC IP address>/redfish/v1/UpdateService -X PATCH -d '{\"HttpPushUriOptions\": {\"HttpPushUriApplyTime\": {\"ApplyTime\": \"Immediate\"}}}'  
$ curl -k -H \"Content-Type: application/octet-stream\" -u <PMC username>:<PMC password> -X POST -T <firmware image path> https://<PMC IP address>/redfish/v1/UpdateService
```

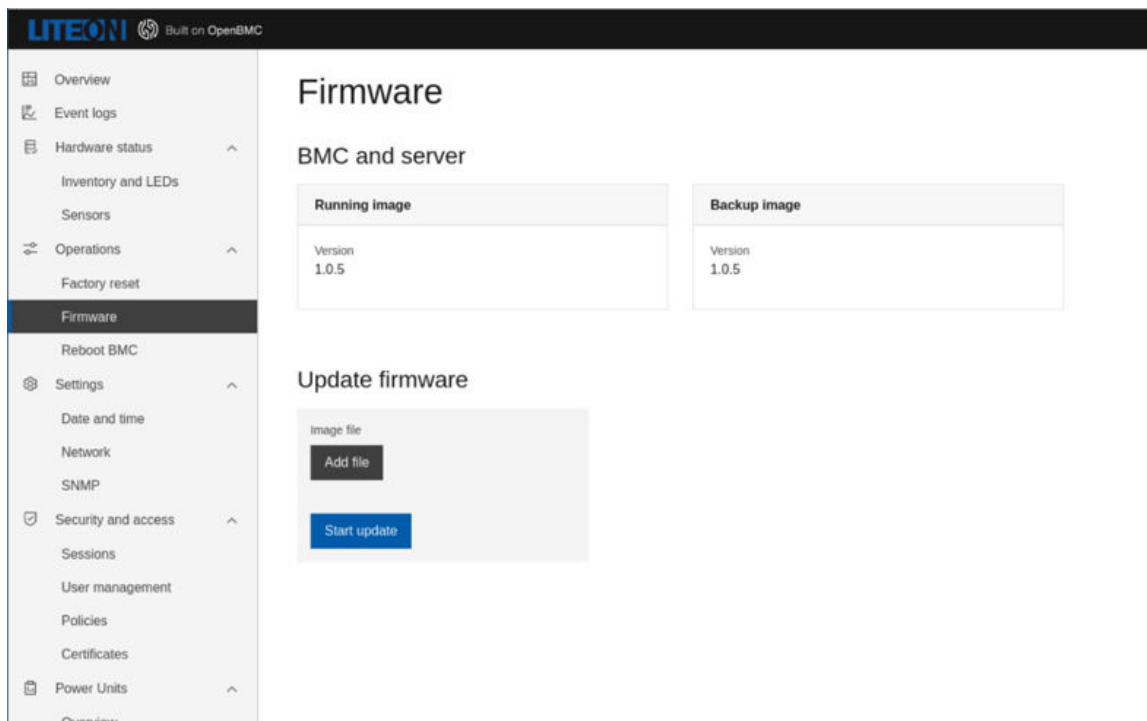
Update the PSU firmware via Redfish CLI commands

Update the Lite-On power shelf PSU firmware using the Linux command `curl` from a Redfish-enabled client system.

Command Syntax: `$ curl -k -H "Content-Type: application/octet-stream" -u <PMC username>:<PMC password> -H "LiteonPowerDeviceId: <PSU device ID>" --data-binary @<firmware image path> -X POST https://<PMC IP address>/redfish/v1/Chassis/chassis/Power/Oem/LiteOn/PowerUnits/Actions/UpdateService/`

Update the firmware via PMC Web GUI

Navigate to the **Firmware** section of the PMC Web GUI and select **Add File** to import the firmware file. Click **Start update** to initiate the process. This procedure will simultaneously update both the **Running image** and **Backup image**.



Deploy the operating system

Use this section to deploy an operating system on the rack solution.

Available operating systems

Ubuntu 24.04 with NVIDIA optimized HWE kernel (kopt)

Complete list of available operating systems: <https://lenovopress.lenovo.com/osig>.

Manual deployment

If you cannot access the above tools, follow the instructions below, download the corresponding OS *Installation Guide*, and deploy the operating system manually by referring to the guide.

1. Go to <https://datacentersupport.lenovo.com/solutions/server-os>.
2. Select an operating system from the navigation pane and click **Resources**.
3. Locate the "OS Install Guides" area and click the installation instructions. Then, follow the instructions to complete the operation system deployment task.

Appendix A. Getting help and technical assistance

If you need help, service, or technical assistance or just want more information about Lenovo products, you will find a wide variety of sources available from Lenovo to assist you.

On the World Wide Web, up-to-date information about Lenovo systems, optional devices, services, and support are available at:

<http://datacentersupport.lenovo.com>

Note: IBM is Lenovo's preferred service provider for ThinkSystem.

Before you call

Before you call, there are several steps that you can take to try and solve the problem yourself. If you decide that you do need to call for assistance, gather the information that will be needed by the service technician to more quickly resolve your problem.

Attempt to resolve the problem yourself

You can solve many problems without outside assistance by following the troubleshooting procedures that Lenovo provides in the online help or in the Lenovo product documentation. The online help also describes the diagnostic tests that you can perform. The documentation for most systems, operating systems, and programs contains troubleshooting procedures and explanations of error messages and error codes. If you suspect a software problem, see the documentation for the operating system or program.

You can find the product documentation for your ThinkSystem products at the following location:

<https://pubs.lenovo.com/>

You can take these steps to try to solve the problem yourself:

- Check all cables to make sure that they are connected.
- Check the power switches to make sure that the system and any optional devices are turned on.
- Check for updated software, firmware, and operating-system device drivers for your Lenovo product. (See the following links) The Lenovo Warranty terms and conditions state that you, the owner of the Lenovo product, are responsible for maintaining and updating all software and firmware for the product (unless it is covered by an additional maintenance contract). Your service technician will request that you upgrade your software and firmware if the problem has a documented solution within a software upgrade.
 - Drivers and software downloads
 - <https://datacentersupport.lenovo.com/products/solutions-and-software/aiinfrastructure/gb300nvl72/7djv/downloads/driver-list/>
 - Operating system support center
 - <https://datacentersupport.lenovo.com/solutions/server-os>
 - Operating system installing instructions
 - <https://pubs.lenovo.com/thinkedge#os-installation>
- If you have installed new hardware or software in your environment, check <https://serverproven.lenovo.com> to make sure that the hardware and software are supported by your product.
- Refer to “Problem Determination” in *User Guide* or *Hardware Maintenance Guide* for instructions on isolating and solving issues.

- Go to <http://datacentersupport.lenovo.com> and check for information to help you solve the problem.

To find the Tech Tips available for your rack solution:

1. Go to <http://datacentersupport.lenovo.com> and navigate to the support page for your rack solution.
2. Click on **How To's** from the navigation pane.
3. Click **Article Type → Solution** from the drop-down menu.

Follow the on-screen instructions to choose the category for the problem that you are having.

- Check Lenovo Data Center Forum at https://forums.lenovo.com/t5/Datacenter-Systems/ct-p/sv_eg to see if someone else has encountered a similar problem.

Gathering information needed to call Support

If you require warranty service for your Lenovo product, the service technicians will be able to assist you more efficiently if you prepare the appropriate information before you call. You can also go to <http://datacentersupport.lenovo.com/warrantylookup> for more information about your product warranty.

Gather the following information to provide to the service technician. This data will help the service technician quickly provide a solution to your problem and ensure that you receive the level of service for which you might have contracted.

- Hardware and Software Maintenance agreement contract numbers, if applicable
- Machine type number (Lenovo 4-digit machine identifier). Machine type number can be found on the ID label, see “[Identify the compute tray and access the BMC](#)” on page 35.
- Model number
- Serial number
- Current system UEFI and firmware levels
- Other pertinent information such as error messages and logs

As an alternative to calling Lenovo Support, you can go to <https://support.lenovo.com/servicerequest> to submit an Electronic Service Request. Submitting an Electronic Service Request will start the process of determining a solution to your problem by making the pertinent information available to the service technicians. The Lenovo service technicians can start working on your solution as soon as you have completed and submitted an Electronic Service Request.

Collecting service data

To clearly identify the root cause of a rack solution issue or at the request of Lenovo Support, you might need collect service data that can be used for further analysis. Service data includes information such as event logs and hardware inventory.

Service data can be collected through the following tools:

- “[Compute tray BMC FFDC logs](#)” on page 46
- “[NVDebug logs](#)” on page 47

Compute tray BMC FFDC logs

1. Navigate to the **Maintenance → Save Server Data** page within the compute tray BMC.
2. Click **Download Server Data** to download FFDC log information for problem escalation. These logs contain the following data for a single compute tray:
 - System inventory information
 - System event logs (SEL)

- Sensor status

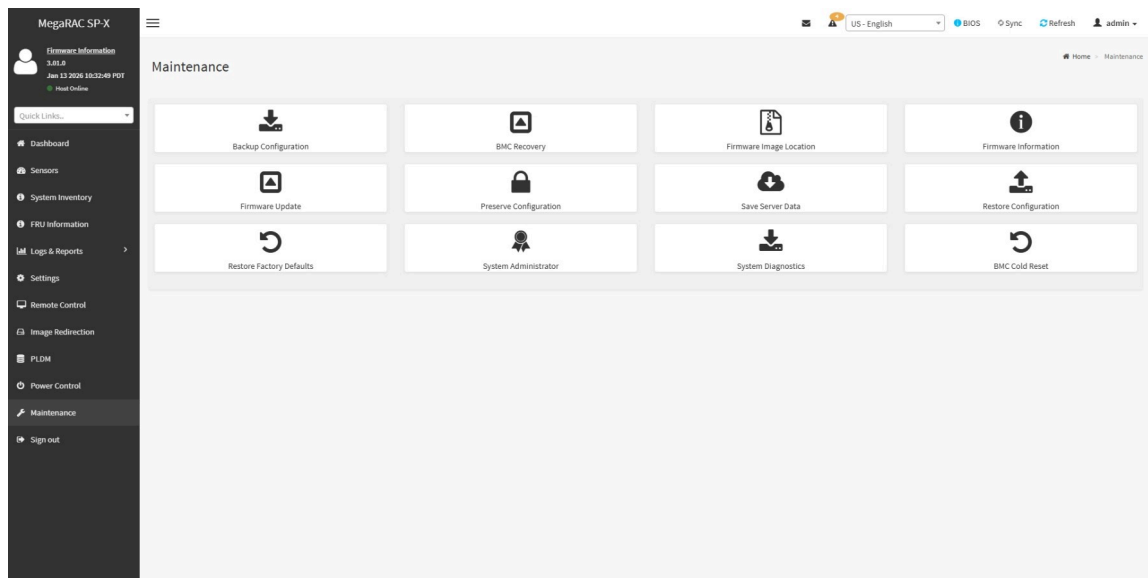


Figure 21. Compute tray BMC FFDC logs

NVDebug logs

Use the NVDebug tool to gather OOB diagnostic logs from one or more compute trays, NVLink switch trays, or power shelves. The tool interfaces with the device BMC to capture data for escalation. Before execution, update the **configuration file** (included in the tool package) with the target device connection information.

For more information about NVDebug, refer the following information:

- https://docs.nvidia.com/multi-node-nvlink-systems/nvdebug-guide/nvdebug_cli.html
- https://docs.nvidia.com/multi-node-nvlink-systems/nvdebug-guide/configuration_guide.html
- https://docs.nvidia.com/multi-node-nvlink-systems/nvdebug-guide/log_collection_guide.html
- https://docs.nvidia.com/multi-node-nvlink-systems/nvdebug-guide/troubleshooting_guide.html

1. Connect a client device via a hub to the RJ-45 OS management port (1) and RJ-45 BMC management port (2) on a compute tray using two cables to establish the tray as the NVDebug host.



2. Download the tool from <https://partners.nvidia.com/> and copy it to the client device.
3. Unzip the package with the following two commands:

```
sudo unzip <NV Debug tool file name>.zip
sudo tar zxvf nvdebug-linux-arm64-<version name>.tar.gz
```

Note: Use the AMD64 for laptop environments.

4. Edit the tool_config.yaml file with the following commands:

```
vim tool_config.yaml
SKIP_BMC_SSH_LOGS: false
```
5. Choose to run the debug tool on either a single compute tray or multiple trays simultaneously:

- Single compute tray:
Execute the following command: `sudo ./nvdebug collect -i <target node BMC IP address> -u <BMC Username> -p <BMC Password> -b "GB300 NVL" -I <target node OS IP address> -U <OS Username> -H <OS Password> -r <BMC Username> -w <BMC Password>`

Note: See the NVDebug User Guide (located in the tool's ZIP file) for complete platform parameter definitions.

- Multiple compute trays simultaneously:
 - Prepare the configuration files first. These files are located in the unzipped tool package folder. Use a text editor to update the following:
 - config.yaml
 - dut_config.yaml
 - In the config.yaml file, set PLATFORM to arm64 and TargetBaseboard to the corresponding device type. Use the following values:
 - GB300 NVL for compute trays
 - GB300 NVSwitchTray for NVLink switch trays
 - PowerShelfController for power shelves

Then, set SKIP_BMC_SSH_LOGS to false.

```
PLATFORM: "arm64"

# Server Baseboard Type
# The accepted values are as follows:
# "Hopper-HGX-8-GPU", "Blackwell-HGX-8-GPU",
# "HGX B300",
# "GH200",
# "C2",
# "GH200 NVL", "GB200 NVL", "GB300 NVL",
# "GB200 NVL NVSwitchTray", "GB300 NVL NVSwitchTray",
# "MGX C2", "MGX-GH200", "MGX-GH200-NVL2",
# "MGX-4U-NVL16", "DC-Blackwell-PCIe", "DC-Hopper-PCIe",
# "PowerShelfController"
#
# When running parse option in NVDebug either baseboard type Hopper-HGX-8-GPU
# or path to FPGA register mapping should be provided under FPGA_REG_MAPPING_
TargetBaseboard: "GB300 NVL"

# Path to user-provided FPGA register mapping file to be used when running --
# If not provided, we use the default file for the passed baseboard type in T
FPGA_REG_MAPPING_FILE: ""

# Automatically parse collected logs and re-zip the results
# When set to true, NVDebug will automatically run the parser on collected lo
# and create a new zip archive with the decoded content
AUTO_PARSE: false

# Set flag to true to remove IP addresses, username and password from plainte
LogSanitization: True

# Skip logs that require port forwarding
# When set to true, all Redfish logs will be skipped when the platform is HGX
SKIP_PORT_FW: false

# Skip SSH logs flag
# When set to true, collection of SSH logs will be skipped
SKIP_BMC_SSH_LOGS: false
```

- In the config.yaml file, update the BMC IP address and credentials for target compute tray.

```

BMC_IP: "192.168.1.70"

# BMC credentials
# This can also be provided via CLI using the -u and -p options respective
BMC_USERNAME: "admin"
BMC_PASSWORD: "adminadmin"

# BMC SSH Credentials
BMC_SSH_USERNAME: "sysadmin"
BMC_SSH_PASSWORD: "superuser"
# BMC_SSH_KEY_PATH: "/root/.ssh"
BMC_SSH_PASSWORDLESS: true
# BMC_SSH_PORT: 22

# HMC Configuration
# HMC_TCP_PORT: 80 # Default is 80, use 443 for HMC with SSL enabled, if pr
# HMC_RF_PROTOCOL: "http" # Default is "http", use "https" for HMC with SSL

# Redfish credentials
# If not provided, it is assumed to be the same as the BMC credentials
RF_User: "admin"
RF_Pass: "adminadmin"

# Port to be used for port forwarding
# To collect certain logs from HMC, port forwarding needs to be setup. NVDe
# If no port is provided below, the default port 18888 will be used
TUNNEL_TCP_PORT: ""
ipmi_cipher: "-c17"

# Host details for Host log collection
# If IP or credentials are not provided, NVDebug assumes it is running on t
HOST_IP: "192.168.1.109"
HOST_USERNAME: "nvidia"
HOST_PASSWORD: "nvidia"

```

- d. Run the following five commands on the host device to set up the environment:


```

sudo apt install ipmitool nvme-cli pciutils dmidecode lshw opensm
sudo apt install nvidia-fabricmanager-580
ssh-keygen -t rsa-b 4096 -f ~/.ssh/nvdebug_key
ssh-copy-id -i ~/.ssh/nvdebug_key.pub nvidia@<host IP address>
ssh-copy-id -i ~/.ssh/nvdebug_key.pub sysadmin@<target node BMC IP address>

```
- e. Run the following two commands on the client device to set up the host environment:


```

sudo apt install ipmitool nvme-cli pciutils dmidecode lshw opensm nvls
sudo apt install nvidia-fabricmanager-580

```
- f. Run the following command to edit the OpenSSH server daemon configuration file: `sudo vim /etc/ssh/sshd_config`

Then, Add the following parameters to the file:

PubkeyAuthentication → yes
 AuthorizedKeysFile → .ssh/authorized_keys

```

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default
AuthorizedKeysFile .ssh/authorized_keys

```

- g. On the host, execute the following command to edit the sudo configuration file: `sudo vim /etc/sudoers`

Add the following entry to the file: `nvidiaALL=(ALL) NOPASSWD:ALL`

```

# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults    env_reset
Defaults    mail_badpass
Defaults    secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
Defaults    use_pty
nvidia ALL=(ALL) NOPASSWD: ALL

```

- h. Execute the NVDebug tool with the following command: `sudo ./nvdebug -i <BMC IP address> -u <username> -p <password> -t arm64 -b "<hardware platform>" -I <host IP address> -U nvidia -H nvidia -o .`

```
soulin@soulin-virtual-machine:~/Desktop/nvdebug_v1.8/nvdebug$ sudo ./nvdebug -i
192.168.1.191 -u admin -p adminadmin -t arm64 -b "GB300 NVL" -I 192.168.1.166 -U
nvidia -H nvidia -o .
Platform and baseboard already configured in config files. Skipping detection.
Log directory created at /home/soulin/Desktop/nvdebug_v1.8/nvdebug/nvdebug_logs_
17_10_2025_14_37_52
Using 4 resources for parallel collection
Starting collection for DUT dut-1
^[[ORdut-1: Platform type: arm64
dut-1: BMC IP: XXXX, Host IP: XXXX
dut-1: Identified system as Model: GB HMC, Partno:699-24764-0001-200, Serialno=1
021725005328
Log collection has started for dut-1
```

Note: Log collection for a single compute tray takes approximately ten minutes to complete. Upon completion, the tool generates a ZIP file within its directory; use this file for problem escalation.

Contacting Support

You can contact Support to obtain help for your issue.

You can receive hardware service through a Lenovo Authorized Service Provider. To locate a service provider authorized by Lenovo to provide warranty service, go to <https://datacentersupport.lenovo.com/serviceprovider> and use filter searching for different countries. For Lenovo support telephone numbers, see <https://datacentersupport.lenovo.com/supportphonenumberlist> for your region support details.

Appendix B. Documents and supports

This section provides handy documents, driver and firmware downloads, and support resources.

Documents download

This section provides introduction and download link for handy documents.

Documents

Download the following product documentations at:

https://pubs.lenovo.com/gb300-nvl72/pdf_files

- **User Guide**

- Complete overview, system configuration, hardware components replacing, and troubleshooting.

Selected chapters from *User Guide*:

- **System Configuration Guide** : Rack solution overview, components identification, system LEDs and diagnostics display, product unboxing, setting up and configuring the rack solution.
- **Hardware Maintenance Guide** : Installing hardware components and troubleshooting.

- **Messages and Codes Reference**

- BMC and UEFI events

- **UEFI User Guide**

- UEFI setting introduction

- **BMC User Guide**

- BMC Web GUI introduction

Support websites

This section provides driver and firmware downloads and support resources.

Support and downloads

- Drivers and Software download website for Lenovo NVIDIA GB300 NVL72
 - <https://datacentersupport.lenovo.com/products/solutions-and-software/aiinfrastructure/gb300nvl72/7djh/downloads/driver-list/>
- Lenovo Data Center Forum
 - https://forums.lenovo.com/t5/Datacenter-Systems/ct-p/sv_eg
- Lenovo Data Center Support for Lenovo NVIDIA GB300 NVL72
 - <https://datacentersupport.lenovo.com/products/solutions-and-software/aiinfrastructure/gb300nvl72/7djh>
- Lenovo License Information Documents
 - <https://datacentersupport.lenovo.com/documents/Invo-eula>
- Lenovo Press website (Product Guides/Datasheets/White papers)
 - <https://lenovopress.lenovo.com/>
- Lenovo Privacy Statement

- <https://www.lenovo.com/privacy>
- Lenovo Product Security Advisories
 - https://datacentersupport.lenovo.com/product_security/home
- Lenovo Product Warranty Plans
 - <http://datacentersupport.lenovo.com/warrantylookup>
- Lenovo Server Operating Systems Support Center website
 - <https://datacentersupport.lenovo.com/solutions/server-os>
- Lenovo ServerProven website (Options compatibility lookup)
 - <https://serverproven.lenovo.com>
- Operating System Installation Instructions
 - <https://pubs.lenovo.com/thinkedge#os-installation>
- Submit an eTicket (service request)
 - <https://support.lenovo.com/servicerequest>
- Subscribe to Lenovo Data Center Group product notifications (Stay up to date on firmware updates)
 - <https://datacentersupport.lenovo.com/solutions/ht509500>

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Important notes

Processor speed indicates the internal clock speed of the processor; other factors also affect application performance.

CD or DVD drive speed is the variable read rate. Actual speeds vary and are often less than the possible maximum.

When referring to processor storage, real and virtual storage, or channel volume, KB stands for 1 024 bytes, MB stands for 1 048 576 bytes, and GB stands for 1 073 741 824 bytes.

When referring to hard disk drive capacity or communications volume, MB stands for 1 000 000 bytes, and GB stands for 1 000 000 000 bytes. Total user-accessible capacity can vary depending on operating environments.

Maximum internal hard disk drive capacities assume the replacement of any standard hard disk drives and population of all hard-disk-drive bays with the largest currently supported drives that are available from Lenovo.

Each solid-state memory cell has an intrinsic, finite number of write cycles that the cell can incur. Therefore, a solid-state device has a maximum number of write cycles that it can be subjected to, expressed as total bytes written (TBW). A device that has exceeded this limit might fail to respond to system-generated commands or might be incapable of being written to. Lenovo is not responsible for replacement of a device that has exceeded its maximum guaranteed number of program/erase cycles, as documented in the Official Published Specifications for the device.

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When you attach a monitor to the equipment, you must use the designated monitor cable and any interference suppression devices that are supplied with the monitor.

Additional electronic emissions notices are available at:

https://pubs.lenovo.com/important_notices/

Taiwan Region BSMI RoHS declaration

單元 Unit	限用物質及其化學符號 Restricted substances and its chemical symbols					
	鉛 Lead (Pb)	汞 Mercury (Hg)	鎘 Cadmium (Cd)	六價鉻 Hexavalent chromium (Cr+6)	多溴聯苯 Polybrominated biphenyls (PBB)	多溴二苯醚 Polybrominated Diphenyl ethers (PBDE)
機箱外殼	—	○	○	○	○	○
機械組件	—	○	○	○	○	○
散熱模組	—	○	○	○	○	○
內存模組	—	○	○	○	○	○
處理器模組	—	○	○	○	○	○
電纜組件	—	○	○	○	○	○
電源供應器	—	○	○	○	○	○
儲存設備	—	○	○	○	○	○
電路板	—	○	○	○	○	○
備考 1. “超出 0.1 wt %” 及 “超出 0.01 wt %” 係指限用物質之百分比含量超出百分比含量基準值。 Note 1 : “Exceeding 0.1 wt %” and “exceeding 0.01 wt %” indicate that the percentage content of the restricted substance exceeds the reference percentage value of presence condition. 備考 2. “○” 係指該項限用物質之百分比含量未超出百分比含量基準值。 Note 2 : “○” indicates that the percentage content of the restricted substance does not exceed the percentage of reference value of presence. 備考 3. “—” 係指該項限用物質為排除項目。 Note 3 : The “—” indicates that the restricted substance corresponds to the exemption.						

Taiwan Region import and export contact information

Contacts are available for Taiwan Region import and export information.

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